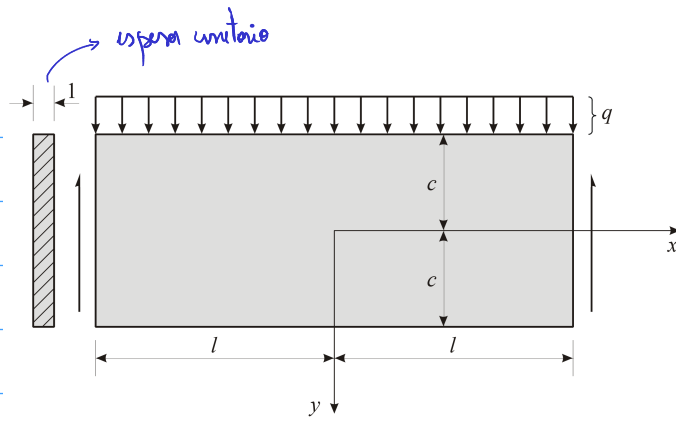




$$\sigma_{xx} = \frac{q}{2l}(l^2 - x^2)y + \frac{q}{2l}\left(\frac{2}{3}y^3 - \frac{2}{5}c^2y\right)$$

$$\sigma_{yy} = -\frac{q}{2l}\left(\frac{1}{3}y^3 - c^2y + \frac{2}{3}c^3\right)$$

$$\sigma_{xy} = -\frac{q}{2l}(c^2 - y^2)x$$

$$\sigma_{zz} = \sigma_{xz} = \sigma_{zy} = 0$$

① Verificar el equilibrio

② Verificar los CBs

③ Verificar la ecuación de compatibilidad de deformaciones

① $\frac{\partial \sigma_{ij}}{\partial x_i} + X_j = 0$ No hay fuerzas de cuerpo

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} = 0 \Rightarrow -\frac{q}{I}xy + \frac{q}{I}xy = 0 \Rightarrow \text{Verificar la ecuación de equil. en direcc. } x$$

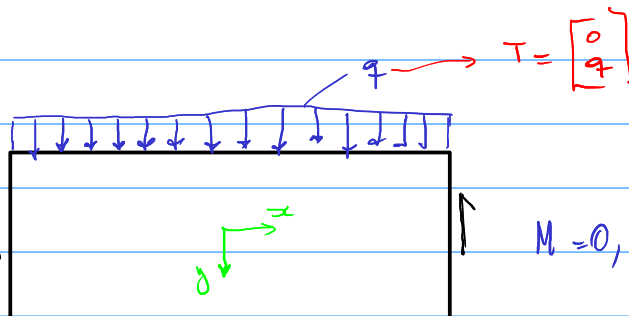
$$\frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{xy}}{\partial x} = 0 \Rightarrow \dots$$

$$\frac{\partial \sigma_{xx}}{\partial x} = \frac{q}{2I}(-2x)y = -\frac{q}{I}xy$$

$$\frac{\partial \sigma_{xy}}{\partial y} = -\frac{q}{2I}(-2y)x = \frac{q}{I}xy$$

2)

Esfuerzos Resultantes
 $M=0, N=0, Q \neq 0$
 momento normal corte

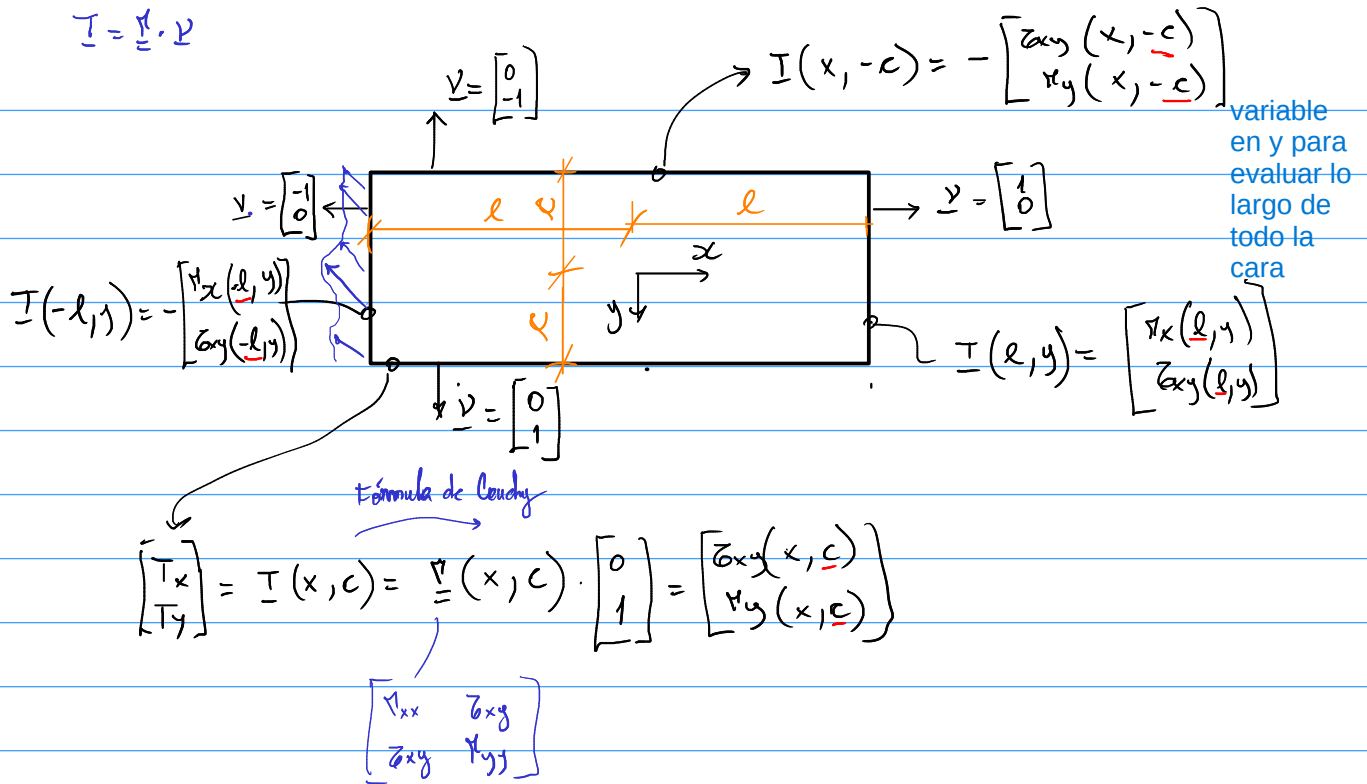


$$M=0, N=0, Q \neq 0$$

$$M=0, N=0, Q=0$$

$$T=0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\underline{T} = \underline{\sigma} \cdot \underline{n}$$



- Para los lados con normal $\pm y$ (lado inferior y superior)

$$\sigma_{xy}(x, \pm c) = -\frac{q}{2I} [c^2 - (\pm c)^2] \cdot x = 0$$

donde aparece y remplazo por c

$$\tau_{yy}(x, \pm c) = -\frac{q}{2I} \left(\frac{1}{3} (\pm c)^3 - c^2 (\pm c) + \frac{2}{3} c^3 \right)$$

$$\begin{aligned} & \rightarrow I = \frac{2c^3}{3} \\ & = -\frac{1}{4} q (\pm 1 \mp 3 + 2) \end{aligned} \begin{cases} \tau_{yy}(x, +c) = 0 \\ \tau_{yy}(x, -c) = -q \end{cases}$$

* lado inferior

$$\underline{T}(x, c) = \begin{bmatrix} \sigma_{xy}(x, c) \\ \tau_{yy}(x, c) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \text{Verifica los CB}$$

* lado superior

$$\underline{T}(x, -c) = -\begin{bmatrix} \sigma_{xy}(x, -c) \\ \tau_{yy}(x, -c) \end{bmatrix} = -\begin{bmatrix} 0 \\ -q \end{bmatrix} = \begin{bmatrix} 0 \\ q \end{bmatrix} \Rightarrow \text{Verif. los CBs}$$

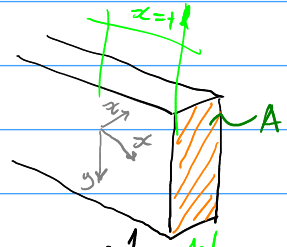
• Para los lados con normal $\pm x$:

$$\tau_x(\pm l, y) = \frac{q}{2I} \left[lz - \cancel{(\pm l)^2} \right] + \frac{q}{2I} \left(\frac{2}{3} y^3 - \frac{2}{5} cz^2 y \right)$$

$$\tau_{xy}(\pm l, y) = -\frac{q}{2I} (cz - y^2) \pm l = \pm \frac{ql}{2I} (cz - y^2)$$

* lado derecho

$$\underline{T}(l, y) = \begin{bmatrix} \tau_x(l, y) \\ \tau_{xy}(l, y) \end{bmatrix}$$



fuerzas resultantes sobre esa cara
N normal
Q de corte

$$\begin{bmatrix} N \\ Q \end{bmatrix} = \iint_A \underline{T}(l, y) \cdot dA = \int_y \int_z \underline{T}(l, y) dz = \int_{-c}^c \underline{T}(l, y) \cdot \underbrace{\int_0^1 1 dz}_{\text{espesor} = 1} dy$$

$$= \begin{bmatrix} \int_{-c}^c \tau_x(l, y) dy \\ \int_{-c}^c \tau_{xy}(l, y) dy \end{bmatrix} = \begin{bmatrix} \int_{-c}^c \tau_x(l, y) \cdot dy \rightarrow \text{LDN} \\ \int_{-c}^c \tau_{xy}(l, y) \cdot dy \rightarrow \text{LDQ} \end{bmatrix}$$

espesor = 1
z constante
siendo z la
profundidad

LDN

$$\int_{-c}^c \tau_x(l, y) \cdot dy = \int_{-c}^c \frac{q}{2I} \left(\frac{2}{3} y^3 - \frac{2}{5} cz^2 y \right) \cdot dy =$$

$$= \frac{q}{2I} \left(\frac{1}{6} y^4 - \frac{1}{5} cz^2 y^2 \right) \Big|_{-c}^c = 0$$

de -c a c para toda la
altura

LDQ

$$\int_{-c}^c \tau_{xy}(l, y) \cdot dy = \int_{-c}^c -\frac{ql}{2I} (cz - y^2) dy = -\frac{ql}{2I} \left(cz \cdot y - \frac{y^3}{3} \right) \Big|_{-c}^c$$

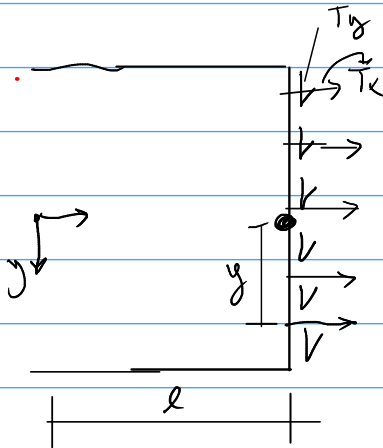
$$= -\frac{ql}{2I} \cdot \frac{4}{3} c^3$$

$$\begin{aligned} & \frac{c^2 c - \cancel{\frac{c^3}{3}} - \left(-c^3 + \cancel{\frac{c^3}{3}} \right)}{\frac{4}{3} c^3} \\ & \frac{c^3 - \cancel{\frac{c^3}{3}} + c^3 - \cancel{\frac{c^3}{3}}}{\frac{4}{3} c^3} \\ & \frac{2c^3 - \frac{2c^3}{3}}{\frac{4}{3} c^3} \\ & 2 \left(\frac{\frac{2}{3} c^3}{\frac{4}{3} c^3} \right) \end{aligned}$$

$$F_D = \begin{bmatrix} N \\ Q \end{bmatrix} = \begin{bmatrix} 0 \\ -ql \end{bmatrix}$$

fuerza hacia arriba

coincide con el sentido de la fuerza dada en el problema



$$M = \iint_A \widetilde{T_x} \cdot dA \cdot y = \iint_A T_x y \cdot dA =$$

$$= \int_{-c}^c \widetilde{V_x}(l, y) \cdot y \, dy = \int_{-c}^c \frac{q}{2I} \left(\frac{2}{3} y^4 - \frac{2}{5} c^2 y^2 \right) dy$$

$$= \frac{q}{2I} \left(\frac{2}{3} \frac{y^5}{5} - \frac{2}{5} c^2 \frac{y^3}{3} \right) \Big|_{-c}^c = 0 \Rightarrow \text{Verificalos}$$

$$\frac{2}{3} \frac{c^5}{5} - \frac{2}{5} \frac{c^5}{3} - \left(\frac{2}{3} \frac{c^5}{5} + \frac{2}{5} \frac{c^5}{3} \right)$$

• dato izquierdo

$$\begin{bmatrix} N \\ Q \end{bmatrix} = \iint_A \underline{T}(-l, y) \cdot d\Delta = - \begin{bmatrix} \int_{-c}^c \tau_x(-l, y) \cdot dy \\ \int_{-c}^c \tau_{xy}(-l, y) \cdot dy \end{bmatrix} =$$

$$= \begin{bmatrix} 0 \\ \int_{-c}^c \underbrace{\tau_{xy}(-l, y)}_{\tau_{xy}(l, y)} \cdot dy \end{bmatrix} = \begin{bmatrix} 0 \\ -ql \end{bmatrix} \Rightarrow \text{Verif. CB}$$

$$M = \iint_A \tau_x(-l, y) \cdot y \cdot d\Delta = \int_{-c}^c \underbrace{\tau_x(-l, y)}_{\tau_x(l, y)} \cdot y \cdot dy = - \underbrace{\int_{-c}^c \tau_x(l, y) \cdot y \cdot dy}_0$$

$$= -0 = 0 \Rightarrow \text{Verif. ca CBs}$$